

ME 333
MECHANICS OF MATERIALS

Lecture 12: Stress-Strain Curves

Ravi Sastri Ayyagari

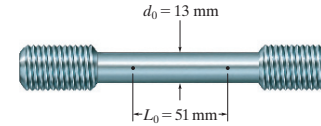
Mechanical Engineering
Indian Institute of Technology Gandhinagar

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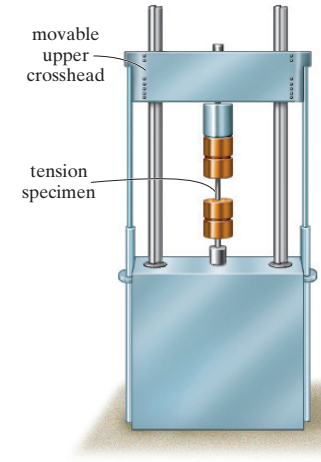
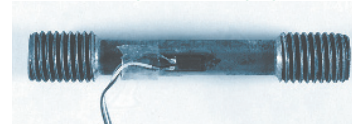
Stress-strain Curves
Mechanical Properties of Materials

Uniaxial Tests

Dogbone Specimen



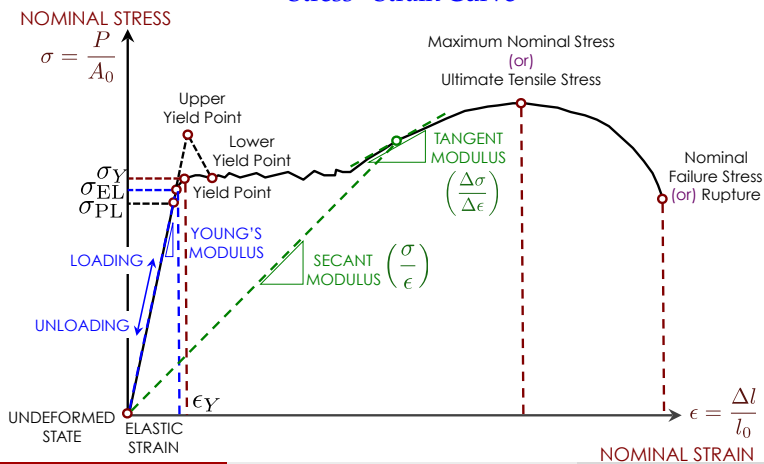
Using a strain gage



Ref for Base Figs: Statics & Mechanics of Materials, R. C. Hibbeler

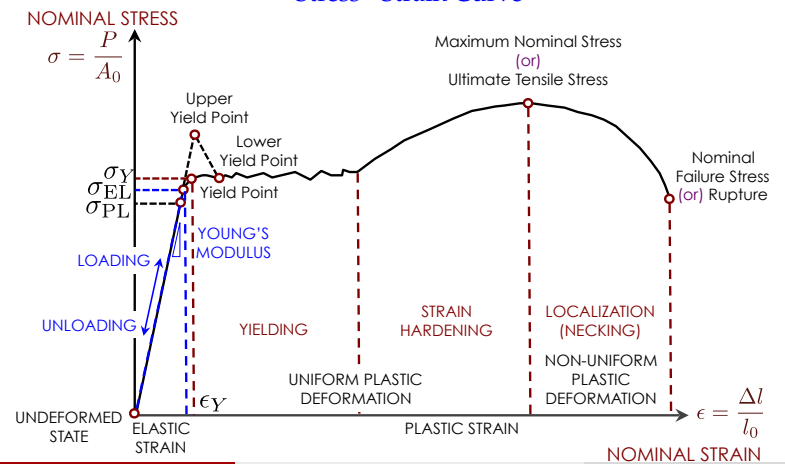
Stress-strain Curves
Mechanical Properties of Materials

Stress - Strain Curve



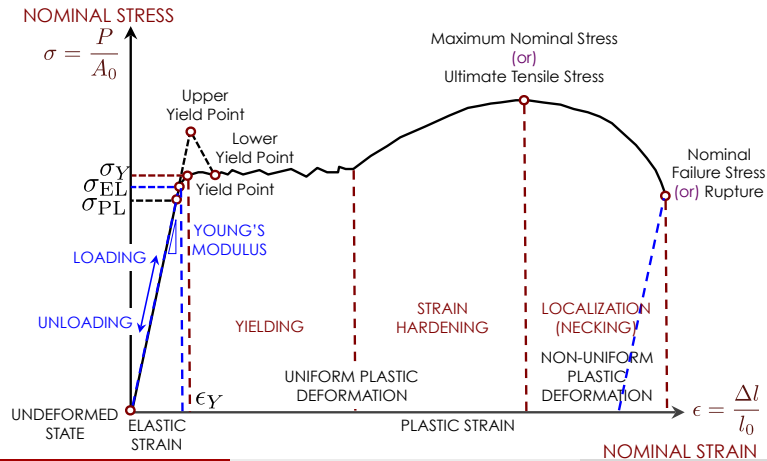
Stress-strain Curves
Mechanical Properties of Materials

Stress - Strain Curve



Mechanical Properties of Materials

Stress - Strain Curve



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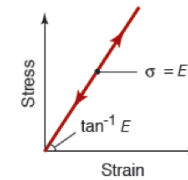
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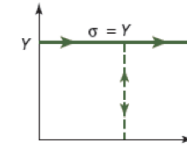
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Mechanical Properties of Materials

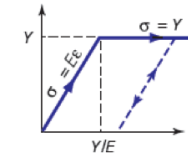
Typical stress - strain curves



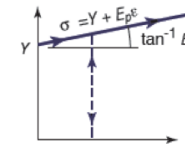
Perfectly Elastic



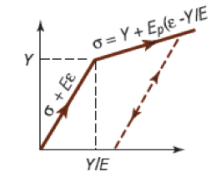
Rigid, Perfectly Plastic



Elastic, Perfectly Plastic



Rigid, Linear Strain Hardening



Elastic, Linear Strain Hardening

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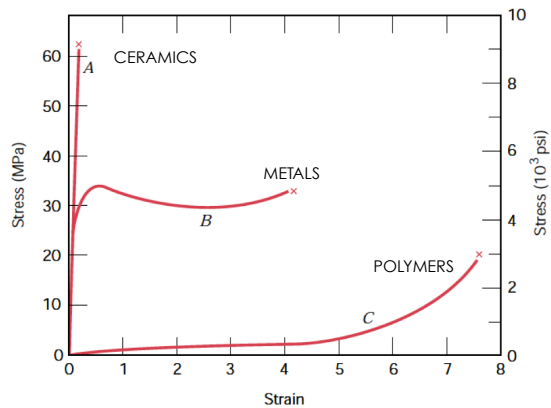
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Mechanical Properties of Materials

Material Types



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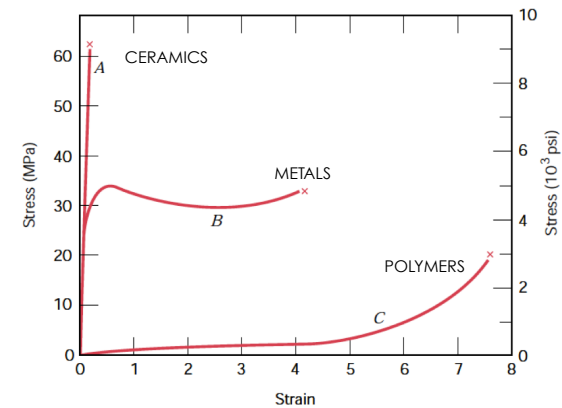
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Mechanical Properties of Materials

Material Types



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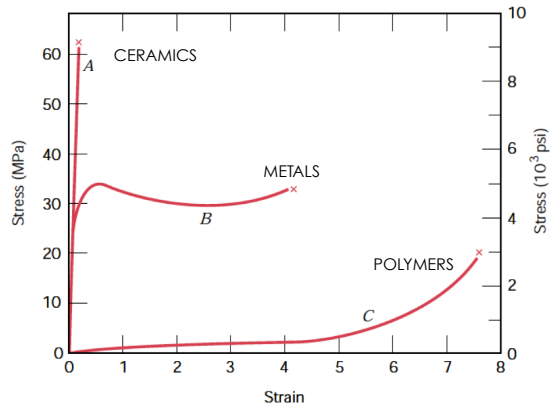
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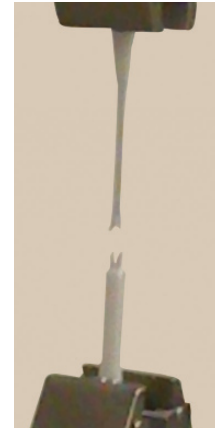
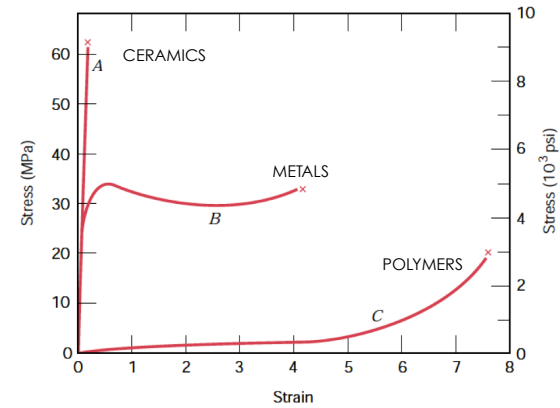
Mechanical Properties of Materials

Material Types



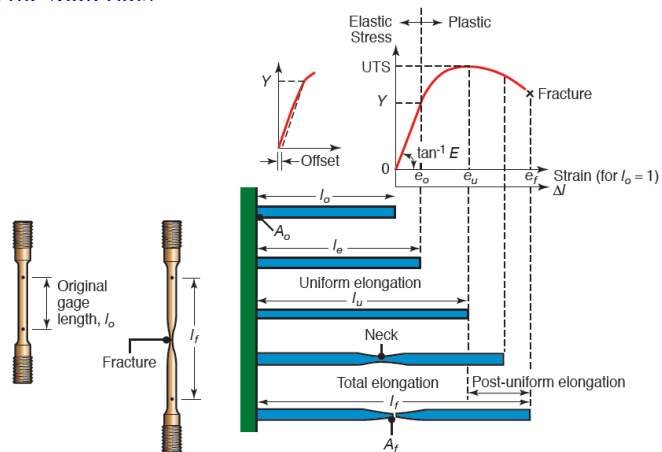
Mechanical Properties of Materials

Material Types



Mechanical Properties of Materials

Necking in Ductile Materials

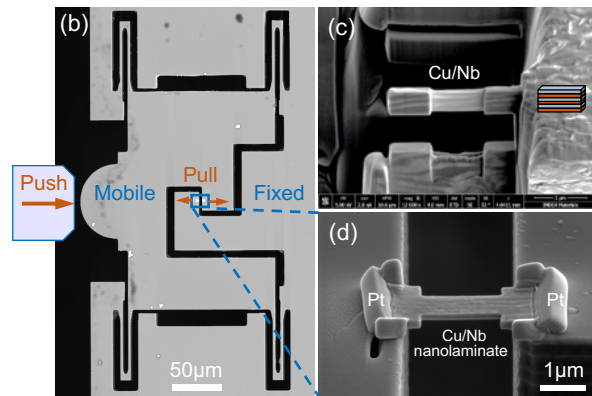


Criterion for Initial Yielding

Q What does yielding actually mean?

Material Yielding

Tensile Test on a Nano-Tensile Sample :



Ref: Liu et. al., Extreme Mechanics Letters (2018)

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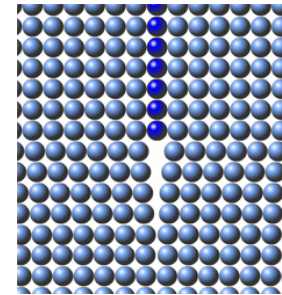
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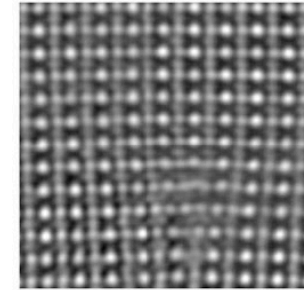
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Concept of Yielding

Dislocation Types :



Edge Dislocation



TEM Image of Edge Dislocation in SrTiO_3

Ref: Jia et. al., PRL 95 (2005)

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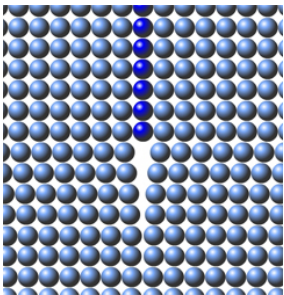
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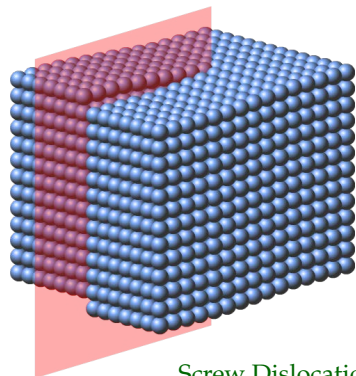
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Concept of Yielding

Dislocation Types :



Edge Dislocation



Screw Dislocation

Ref: Jia et. al., PRL 95 (2005)

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Concept of Yielding

Dislocations & Shear stresses

- Dislocation are driven by shear stresses
- Plastic deformation as a consequence of dislocation activity is therefore driven by shear stresses
- There can be millions of crystals/grains and millions of kilometers of dislocations in a cubic millimeter of plastically deformed metal

Ref: Jia et. al., PRL 95 (2005)

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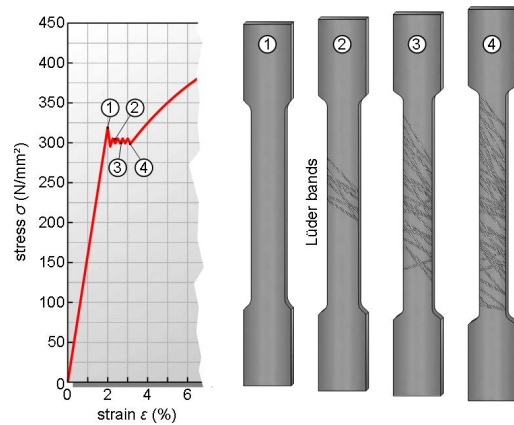
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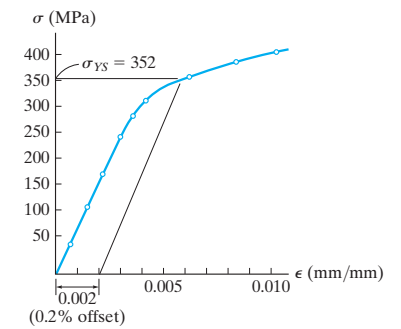
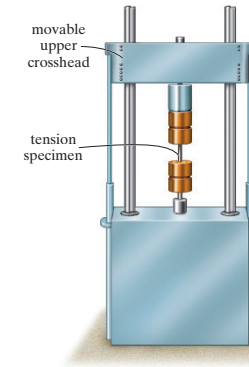
Concept of Yielding

Onset of Plastic deformation - Yielding :



Criterion for Initial Yielding

Uniaxial Loading Scenario :



Al alloy material response

- Yielding is pre-dominantly driven by shear components

Ref: Mechanics of Materials, R. C. Hibbeler

Criterion for Initial Yielding

Q What is the need for a yield criterion - if a simple uniaxial test can give you the yield strength?

Reading Assignment

Chapter 4: Section 4.1 - 4.3